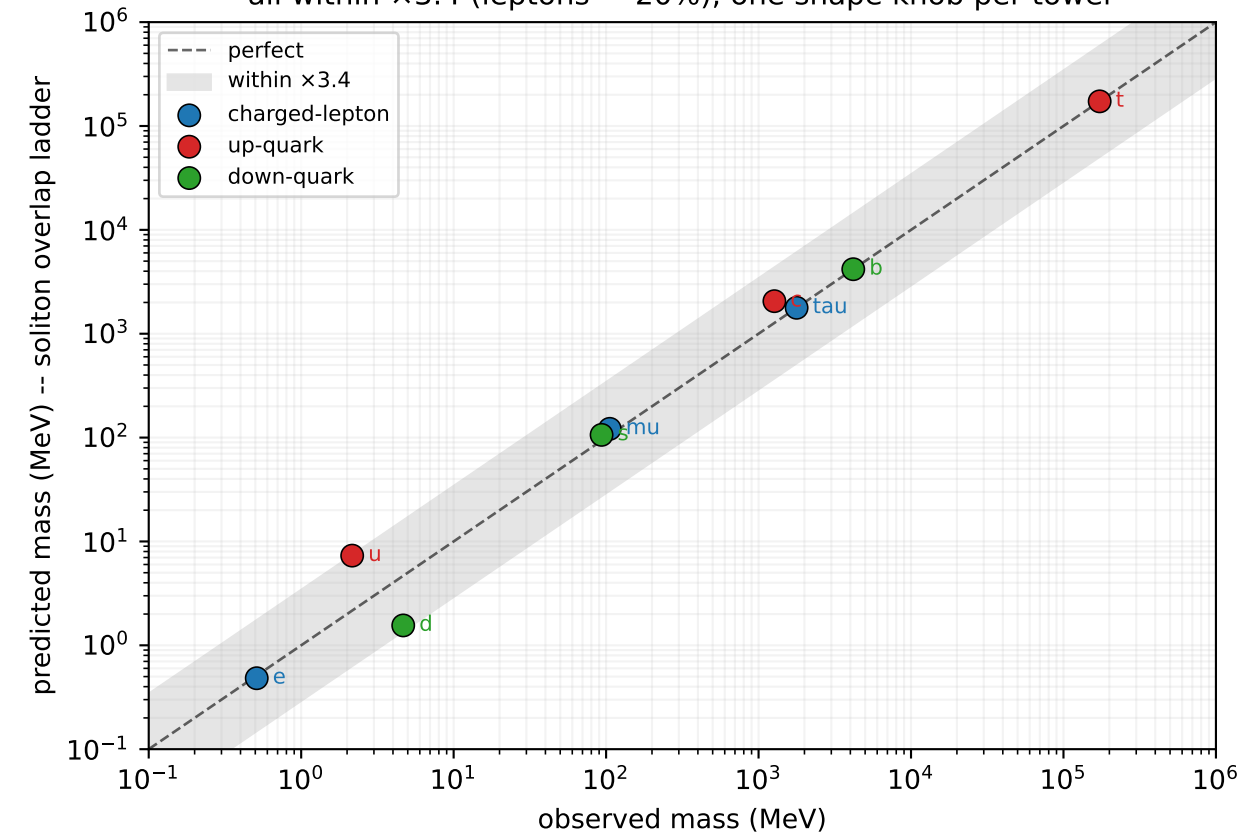


Ab initio (best effort): the 12 fermion masses as soliton overlap integrals -- roughly right, exact numbers owed to lattice (L4)

9 charged fermions from the overlap ladder
all within $\times 3.4$ (leptons $\sim 20\%$), one shape knob per tower



The four towers: solid=observed, dashed=ladder
hierarchy by condensate coupling; neutrinos at the floor

